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B.Tech. Degree IV Semester Examination April 2019

IT 15-1403 SYSTEM PROGRAMMING (2015 Scheme)

Time : 3 Hours

Maximum Marks : 60

PART A (Answer ALL questions)

(10 × 2 = 20)

- I. (a) Explain the functions of two passes of a simple assembler.
- (b) What are the different records in an object program format?
- (c) How does the assembler handle Literals? What is the difference between immediate operands and literals?
- (d) Explain the terms loading, relocation and linking.
- (e) Differentiate Linking loader and Linkage editor with a neat diagram.
- (f) What is concatenation of macro parameters? Give examples.
- (g) Explain the generation of unique labels in macro with example.
- (h) Explain how the Keyword macro parameters work in the design of macro processors.
- (i) Explain basic functions of an operating system.
- (j) Write a short note on virtual machines.

PART B

(4 × 10 = 40)

- II. Write the algorithm for a 2 pass assembler with the necessary data structures needed. (10)
- OR**
- III. (a) What are assembler directives? Explain with example. (3)
- (b) Write a short note on control sections and program linking. (7)
- IV. (a) Explain any two methods for specifying relocation as a part of object program with suitable algorithms. (6)
- (b) Differentiate absolute loader and boot strap loader. (4)
- OR**
- V. Explain the algorithm for a linking loader with data structures. (10)
- VI. (a) What is meant by recursive macro? How is it implemented in one pass macro processor? (5)
- (b) Explain conditional macro expansion. (5)
- OR**
- VII. What are the different data structures used in the implementation of a single pass macro processor? Write the algorithm for a single pass macro processor. (10)
- VIII. Explain multiprocessor operating systems and distributed operating systems with a neat diagram. (10)
- OR**
- IX. Explain : (i) User interface and run time environment. (10)
- (ii) Hierarchical structure of an operating system.